BEng (Hons) Software Engineering

L4 – Semester1

Module: 4COSC001W: Software Development I

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Type Of Assignment: Individual Coursework

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Table of Contents

List of Figures

List of Tables

1. Pseudocode

```
BEGIN
   FUNCTION validate_date_input(prompt, minValue, maxValue)
      WHILE True DO
         user_input = INPUT(prompt)
         TRY
            value = CONVERT user_input TO INTEGER
            IF minValue IS NOT None AND value < minValue THEN
               PRINT 'Out of range - values must be in the range minValue and maxValue.'
               CONTINUE
            ENDIF
            IF maxValue IS NOT None AND value > maxValue THEN
               PRINT 'Out of range - values must be in the range minValue and maxValue.'
               CONTINUE
            ENDIF
            RETURN value
         EXCEPT ValueError
            PRINT 'Integer required'
         ENDTRY
      ENDWHILE
   END FUNCTION

   FUNCTION validate_continue_input()
      WHILE True DO
         continue_program = INPUT("Do you want to select another data file for a different date? (Y/N): ").UPPER()
         IF continue_program == "Y" THEN
            RETURN True
         ELSE IF continue_program == "N" THEN
```

```
            PRINT "End of run"
            RETURN False
        ELSE
            PRINT 'Invalid: Please enter “Y” or “N”'
        ENDIF
    ENDWHILE
END FUNCTION

FUNCTION process_csv_data(file_name)
    INITIALIZE outcomes and traffic_data variables
    TRY
        OPEN file_name IN read mode
        READ csv_data USING DictReader
        FOR EACH row IN csv_data DO
            INCREMENT total_vehicles
            EXTRACT vehicle_type, electricHybrid, junction_name, and other fields
            IF vehicle_type == "truck" THEN
                INCREMENT total_trucks
            ENDIF
            IF electricHybrid == "true" THEN
                INCREMENT electric_vehicles
            ENDIF
            IF vehicle_type IN ["bicycle", "motorcycle", "scooter"] THEN
                INCREMENT twoWheeled_vehicles
            ENDIF
            IF junction_name == "elm avenue/rabbit road" AND vehicle_type == "bus" AND
heading_out == "n" THEN
                INCREMENT busses_leaving_elmAvenue_heading_north
            ENDIF
            IF heading_in == heading_out THEN
```

```
            INCREMENT vehicles_passing_without_turning
        ENDIF
        CALCULATE truck_percentage
        CALCULATE average_bicycle_perhour
        IF vehicle_speed > junction_speed THEN
            INCREMENT vehicles_over_speedlimit
        ENDIF
        IF junction_name == "elm avenue/rabbit road" THEN
            INCREMENT vehicles_through_elmAvenue
        ENDIF
        IF junction_name == "hanley highway/westway" THEN
            INCREMENT vehicles_through_hanleyhigh
            INCREMENT vehicles_inPeak_hanleyhigh FOR corresponding hour
        ENDIF
        UPDATE peak_hour_format AND rain_hours
    ENDFOR
    STORE all metrics IN outcomes
    RETURN outcomes
  CATCH FileNotFoundError
    PRINT "File not found"
    RETURN None
  ENDTRY
END FUNCTION

FUNCTION display_outcomes(outcomes)
  PRINT each metric in outcomes WITH labels
END FUNCTION

FUNCTION save_results_to_file(outcomes, file_name="results.txt")
  TRY
```

```
        OPEN file_name IN append mode
        WRITE outcomes TO file
    CATCH IOError
        PRINT "Unable to write to the file"
    ENDTRY
END FUNCTION


FUNCTION HistogramApp(traffic_data, date)
    SET traffic_data for two junctions "Elm Avenue/Rabbit Road" and "Hanley
Highway/Westway"
    FORMAT date as DD/MM/YYYY
    INITIALIZE Tkinter window with title "Histogram" and size "1400x600"
    DEFINE colors for the junctions
    INITIALIZE canvas as None
  END FUNCTION

    FUNCTION setup_window()
        CREATE canvas for drawing histogram
        CALL draw_histogram()
    END FUNCTION


    FUNCTION draw_histogram()
        SET constants for bar width, gap, X/Y positions, max value scaling
        DRAW title and axes on canvas
        LOOP over 24 hours to draw bars for traffic data
            CALCULATE bar height based on max value
            DRAW bars for each junction with corresponding color
            DISPLAY values on top of bars if they are non-zero
        CALL draw_legend()
    END FUNCTION
```

```
    FUNCTION draw_legend()
        DRAW legend with colored rectangles for each junction name

    FUNCTION run()
        CALL setup_window()
        START Tkinter main loop to display window
END FUNCTION

FUNCTION MultiCSVProcessor()
    INITIALIZE current_data as None
    FUNCTION load_csv_file(file_path)
        TRY
            CALL process_csv_data(file_path)
            IF data loaded successfully THEN
                STORE data IN current_data
                RETURN True
            ELSE
                PRINT "File not found"
                RETURN False
        CATCH FileNotFoundError
            PRINT "File not found"
            RETURN False
        ENDTRY
   END FUNCTION

    FUNCTION clear_previous_data()
        CLEAR current_data by setting it to None
    END FUNCTION
    FUNCTION handle_user_interaction()
```

```
        WHILE True DO
            day = CALL validate_date_input('Please enter the day (DD): ', 1, 31)
            month = CALL validate_date_input('Please enter the month (MM): ', 1, 12)
            year = CALL validate_date_input('Please enter the year (YYYY): ', 2000, 2024)
            date = FORMAT date as DDMMYYYY
            file_name = CONCAT "traffic_data", date, ".csv"
            PRINT "Selected file:", file_name

            outcomes = CALL load_csv_file(file_name)
            IF outcomes IS None THEN
                CONTINUE
            ENDIF
            CALL display_outcomes(outcomes)
            CALL save_results_to_file(outcomes)
            CALL HistogramApp(traffic_data, date).run()

            IF NOT CALL validate_continue_input() THEN
                BREAK
            ENDIF
        ENDWHILE
    END FUNCTION
    FUNCTION process_files()
        CALL handle_user_interaction()
  END FUNCTION

  FUNCTION main()
    CREATE instance of MultiCSVProcessor
    CALL process_files()
  END FUNCTION
  CALL main()
END
```

2. Test Cases

2.1 Table of Test cases – Validating user Inputs

Test case Description	Input	Expected outcome	Actual outcome	Result
1. Valid Input	Day: 15 Month: 06 Year: 2024	constructs the file name as 'traffic_data15062024.csv'	constructs the file name as 'traffic_data15062024.csv'	PASS
2. Invalid Day Input	Day: 35	Error message: Out of range - values must be in the range 1 and 31. Program prompts for re-entry.	Error message: Out of range - values must be in the range 1 and 31. Program prompts for re-entry.	PASS
3. Invalid Month Input	Month: 13	Error message: Out of range - values must be in the range 1 and 12. Program prompts for re-entry.	Error message: Out of range - values must be in the range 1 and 12. Program prompts for re-entry.	PASS
4. Invalid Year Input	Year: 1995	Error message: Out of range - values must be in the range 2000 and 2024. Program prompts for re-entry.	Error message: Out of range - values must be in the range 2000 and 2024. Program prompts for re-entry.	PASS
5. Non-Numeric Input	Day: abc	Error message: Integer required. Program prompts for re-entry.	Error message: Integer required. Program prompts for re-entry.	PASS
6. Empty Input	Day:	Error message: Integer required. Program prompts for re-entry.	Error message: Integer required. Program prompts for re-entry.	PASS
7. User Continues	'Y' after processing the first file.	Program loops back to the start for new file selection.	Program loops back to the start for new file selection.	PASS
8. User Exits	'N' after processing a file.	Program ends with a message: End of run.	Program ends with a message: End of run.	PASS
9. Invalid Input for User continues	'A' after processing a file	Error message: Invalid: Please enter "Y" or "N"	Error message: Invalid: Please enter "Y" or "N"	PASS

Table 1 Test Cases – Validating user inputs

2.2 Test Screenshots – validating user inputs

```
Python 3.12.1 (tags/v3.12.1:2305ca5, Dec  7 2023, 22:03:25) [MSC v.1937 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Dehansa Dissanayake\Desktop\IIT\DEGREE\SD 1\Coursework\w2119977\code.py
Please enter the day of the survey in the format dd: 15
Please enter the month of the survey in the format MM: 6
Please enter the year of the survey in the format YYYY: 2024

****************************************
data file Selected is: traffic_data15062024.csv
****************************************
The total number of vehicles recorded to this date is 1037
The total number of trucks recorded to this date is 109
The total number of electric_vehicles recorded to this date is 368
The total number of twoWheeled_vehicles recorded to this date is 401
The total number of Busses leaving Elm Avenue/Rabbit Road heading North is 15
The total number of Vehicles through both junctions not turning left or right is 363
The percentage of total vehicles recorded that are trucks for this date is 11%
the average number of Bikes per hour for this date is 7
The total number of Vehicles recorded as over the speed limit for this date is 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is 494
The total number of vehicles recorded through Hanley Highway/Westway junction is 543
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters
The highest number of vehicles in an hour on Hanley Highway/Westway is 39
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is 0

Do you want to select another data file for a different date? (Y/N):
```

Figure 3 Test case 1 - Validating User Inputs

```
Please enter the day of the survey in the format dd: 35
Out of range - values must be in the range 1 and 31.
Please enter the day of the survey in the format dd:
```

Figure 2 Test case 2 - Validating User Inputs

```
Please enter the month of the survey in the format MM: 13
Out of range - values must be in the range 1 and 12.
Please enter the month of the survey in the format MM:
```

Figure 1 Test case 3 - Validating User Inputs

```
Please enter the year of the survey in the format YYYY: 1995
Out of range - values must be in the range 2000 and 2024.
Please enter the year of the survey in the format YYYY:
```

Figure 4 Test case 4 - Validating User Inputs

```
Please enter the day of the survey in the format dd: abc
Integer required
Please enter the day of the survey in the format dd:
```

Figure 5 Test case 5 - Validating User Inputs

```
Please enter the day of the survey in the format dd:
Integer required
Please enter the day of the survey in the format dd:
```

Figure 6 Test case 6 - Validating User Inputs

```
Do you want to select another data file for a different date? (Y/N): Y
Please enter the day of the survey in the format dd:
```

Figure 7 Test case 7 - Validating User Inputs

```
Do you want to select another data file for a different date? (Y/N): N
End of run
>>>
```

Figure 9 Test case 8 - Validating User Inputs

```
Do you want to select another data file for a different date? (Y/N): A
Invalid: Please enter “Y” or “N”
Do you want to select another data file for a different date? (Y/N):
```

Figure 8 Test case 9 - Validating User Inputs

2.3 Table of Test cases – File Handling Tests

No.	Input in the console	Expected outcome	Actual outcome	Result
1. File Exists	Day: 15 Month: 06 Year: 2024 File name: traffic_data15062024.csv created	traffic_data15062023.csv exists. File is processed, and outcomes are displayed and saved.	traffic_data15062023.csv exists. File is processed, and outcomes are displayed and saved.	PASS
2. File Not Found	Day: 15 Month: 03 Year: 2023 File name: traffic_data15032023.csv created	traffic_data15032023.csv (does not exist). Error message: File traffic_data15032023.csv not Found. displayed	traffic_data15032023.csv (does not exist). Error message: File traffic_data15032023.csv not Found. displayed	PASS
3. saving results to a text file	Day: 15 Month: 06 Year: 2024 File name: traffic_data15062024.csv created	Saving results to results.txt	Saving results to results.txt	PASS
4. Appending to text file	Day: 16 Month: 06 Year: 2024 File name: traffic_data16062024.csv created	results.txt appends new data for each file run, maintaining previous data.	results.txt appends new data for each file run, maintaining previous data.	PASS

Table 2 Test Cases - File Handling Test

2.4 Table of Test cases – Histogram Tests

Table 3 Test Cases Histogram

No.	Input in the console	Expected outcome	Actual outcome	Result
1. File Exists	Day: 15 Month: 06 Year: 2024	Process the data and Display the histogram	Process the data and Display the histogram	PASS
2. File Not Found	Day: 15 Month: 03 Year: 2023	Error message: File traffic_data15032023.csv not Found. Displayed in the console and no Histogram is displayed	Error message: File traffic_data15032023.csv not Found. Displayed in the console and no Histogram is displayed	PASS

2.5 Test Screenshots File Handling Tests

```
Python 3.12.1 (tags/v3.12.1:2305ca5, Dec  7 2023, 22:03:25) [MSC v.1937 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Dehansa Dissanayake\Desktop\IIT\DEGREE\SD 1\Coursework\w2119977\code.py
Please enter the day of the survey in the format dd: 15
Please enter the month of the survey in the format MM: 6
Please enter the year of the survey in the format YYYY: 2024

****************************************
data file Selected is: traffic_data15062024.csv
****************************************
The total number of vehicles recorded to this date is 1037
The total number of trucks recorded to this date is 109
The total number of electric_vehicles recorded to this date is 368
The total number of twoWheeled_vehicles recorded to this date is 401
The total number of Busses leaving Elm Avenue/Rabbit Road heading North is 15
The total number of Vehicles through both junctions not turning left or right is 363
The percentage of total vehicles recorded that are trucks for this date is 11%
the average number of Bikes per hour for this date is 7
The total number of Vehicles recorded as over the speed limit for this date is 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is 494
The total number of vehicles recorded through Hanley Highway/Westway junction is 543
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters
The highest number of vehicles in an hour on Hanley Highway/Westway is 39
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is 0

Do you want to select another data file for a different date? (Y/N):
```

Figure 10 Test case 1 – File Handling Tests

```
Please enter the day of the survey in the format dd: 15
Please enter the month of the survey in the format MM: 3
Please enter the year of the survey in the format YYYY: 2023

****************************************
data file Selected is: traffic_data15032023.csv
****************************************
File traffic_data15032023.csv not Found
Please enter the day of the survey in the format dd:
```

Figure 11 Test case 2 – File Handling Tests

```
data file Selected is: traffic_data15062024.csv
The total number of vehicles recorded to this date is 1037
The total number of trucks recorded to this date is 109
The total number of electric vehicles recorded to this date is 368
The total number of two-wheeled vehicles recorded to this date is 401
The total number of Busses leaving Elm Avenue/Rabbit Road heading North is 15
The total number of Vehicles through both junctions not turning left or right is 363
The percentage of total vehicles recorded that are trucks for this date is 11%
the average number of Bikes per hour for this date is 7
The total number of Vehicles recorded as over the speed limit for this date is 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is 494
The total number of vehicles recorded through Hanley Highway/Westway junction is 543
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters
The highest number of vehicles in an hour on Hanley Highway/Westway is 39
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is 0

*******************************

data file Selected is: traffic_data16062024.csv
The total number of vehicles recorded to this date is 101
The total number of trucks recorded to this date is 11
The total number of electric vehicles recorded to this date is 29
The total number of two-wheeled vehicles recorded to this date is 29
The total number of Busses leaving Elm Avenue/Rabbit Road heading North is 0
The total number of Vehicles through both junctions not turning left or right is 38
The percentage of total vehicles recorded that are trucks for this date is 11%
the average number of Bikes per hour for this date is 0
The total number of Vehicles recorded as over the speed limit for this date is 20
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is 52
The total number of vehicles recorded through Hanley Highway/Westway junction is 49
5% of vehicles recorded through Elm Avenue/Rabbit Road are scooters
The highest number of vehicles in an hour on Hanley Highway/Westway is 5
The most vehicles through Hanley Highway/Westway were recorded between 1:00 and 2:00
The number of hours of rain for this date is 3

*******************************
```

Figure 12 Test case 3 and 4 – File Handling Tests

3. Screenshots in full program

```
>>>
= RESTART: C:\Users\Dehansa Dissanayake\Desktop\IIT\DEGREE\SD 1\Coursework\w2119977\code.py
Please enter the day of the survey in the format dd: abc
Integer required
Please enter the day of the survey in the format dd: 35
Out of range - values must be in the range 1 and 31.
Please enter the day of the survey in the format dd: 15
Please enter the month of the survey in the format MM: 13
Out of range - values must be in the range 1 and 12.
Please enter the month of the survey in the format MM: 6
Please enter the year of the survey in the format YYYY: 1995
Out of range - values must be in the range 2000 and 2024.
Please enter the year of the survey in the format YYYY: 2024

****************************************
data file Selected is: traffic_data15062024.csv
****************************************
The total number of vehicles recorded to this date is 1037
The total number of trucks recorded to this date is 109
The total number of electric_vehicles recorded to this date is 368
The total number of twoWheeled_vehicles recorded to this date is 401
The total number of Busses leaving Elm Avenue/Rabbit Road heading North is 15
The total number of Vehicles through both junctions not turning left or right is 363
The percentage of total vehicles recorded that are trucks for this date is 11%
the average number of Bikes per hour for this date is 7
The total number of Vehicles recorded as over the speed limit for this date is 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is 494
The total number of vehicles recorded through Hanley Highway/Westway junction is 543
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters
The highest number of vehicles in an hour on Hanley Highway/Westway is 39
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is 0

Do you want to select another data file for a different date? (Y/N): y
Please enter the day of the survey in the format dd: 15
Please enter the month of the survey in the format MM: 3
Please enter the year of the survey in the format YYYY: 2023

****************************************
data file Selected is: traffic_data15032023.csv
****************************************
File traffic_data15032023.csv not Found
Please enter the day of the survey in the format dd: 16
Please enter the month of the survey in the format MM: 6
Please enter the year of the survey in the format YYYY: 2024

****************************************
data file Selected is: traffic_data16062024.csv
****************************************
The total number of vehicles recorded to this date is 101
The total number of trucks recorded to this date is 11
The total number of electric_vehicles recorded to this date is 29
The total number of twoWheeled_vehicles recorded to this date is 29
The total number of Busses leaving Elm Avenue/Rabbit Road heading North is 0
The total number of Vehicles through both junctions not turning left or right is 38
The percentage of total vehicles recorded that are trucks for this date is 11%
the average number of Bikes per hour for this date is 0
The total number of Vehicles recorded as over the speed limit for this date is 20
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is 52
The total number of vehicles recorded through Hanley Highway/Westway junction is 49
5% of vehicles recorded through Elm Avenue/Rabbit Road are scooters
The highest number of vehicles in an hour on Hanley Highway/Westway is 5
The most vehicles through Hanley Highway/Westway were recorded between 1:00 and 2:00
The number of hours of rain for this date is 3

Do you want to select another data file for a different date? (Y/N): n
End of run
>>>
```

Figure 13 Full running program

4. Screenshots in results.txt

```
results.txt
File    Edit    View

data file Selected is: traffic_data15062024.csv
The total number of vehicles recorded to this date is 1037
The total number of trucks recorded to this date is 109
The total number of electric vehicles recorded to this date is 368
The total number of two-wheeled vehicles recorded to this date is 401
The total number of Busses leaving Elm Avenue/Rabbit Road heading North is 15
The total number of Vehicles through both junctions not turning left or right is 363
The percentage of total vehicles recorded that are trucks for this date is 11%
the average number of Bikes per hour for this date is 7
The total number of Vehicles recorded as over the speed limit for this date is 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is 494
The total number of vehicles recorded through Hanley Highway/Westway junction is 543
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters
The highest number of vehicles in an hour on Hanley Highway/Westway is 39
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is 0

*******************************

data file Selected is: traffic_data16062024.csv
The total number of vehicles recorded to this date is 101
The total number of trucks recorded to this date is 11
The total number of electric vehicles recorded to this date is 29
The total number of two-wheeled vehicles recorded to this date is 29
The total number of Busses leaving Elm Avenue/Rabbit Road heading North is 0
The total number of Vehicles through both junctions not turning left or right is 38
The percentage of total vehicles recorded that are trucks for this date is 11%
the average number of Bikes per hour for this date is 0
The total number of Vehicles recorded as over the speed limit for this date is 20
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is 52
The total number of vehicles recorded through Hanley Highway/Westway junction is 49
5% of vehicles recorded through Elm Avenue/Rabbit Road are scooters
The highest number of vehicles in an hour on Hanley Highway/Westway is 5
The most vehicles through Hanley Highway/Westway were recorded between 1:00 and 2:00
The number of hours of rain for this date is 3

*******************************
```

Figure 14 Screenshots in results.txt

```
data file Selected is: traffic_data21062024.csv
The total number of vehicles recorded to this date is 1334
The total number of trucks recorded to this date is 138
The total number of electric vehicles recorded to this date is 442
The total number of two-wheeled vehicles recorded to this date is 503
The total number of Busses leaving Elm Avenue/Rabbit Road heading North is 19
The total number of Vehicles through both junctions not turning left or right is 494
The percentage of total vehicles recorded that are trucks for this date is 10%
the average number of Bikes per hour for this date is 10
The total number of Vehicles recorded as over the speed limit for this date is 250
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is 651
The total number of vehicles recorded through Hanley Highway/Westway junction is 683
9% of vehicles recorded through Elm Avenue/Rabbit Road are scooters
The highest number of vehicles in an hour on Hanley Highway/Westway is 71
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is 6

*******************************
```

Figure 15Screenshots in results.txt

5. Screenshots Histogram

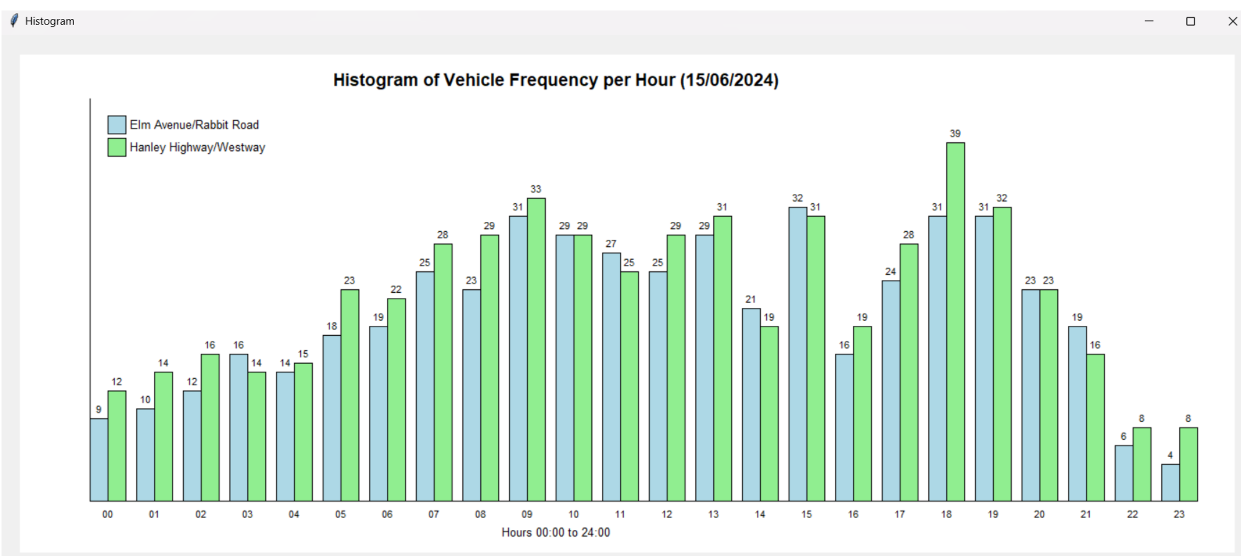

Figure 16 Screenshots in Histogram

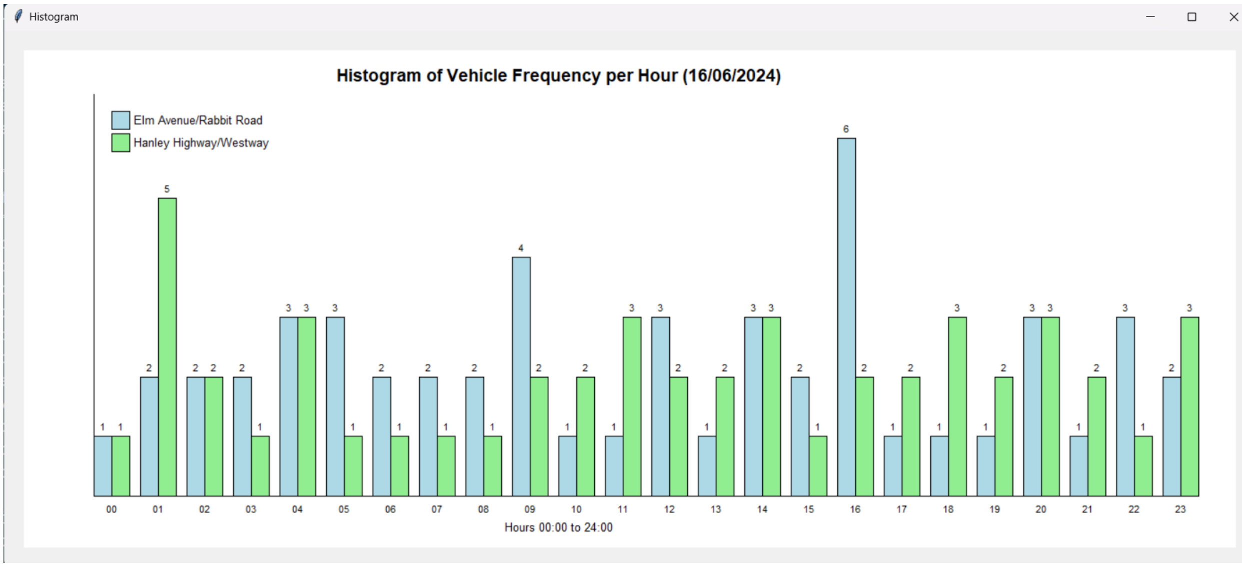

Figure 17 Screenshots in Histogram

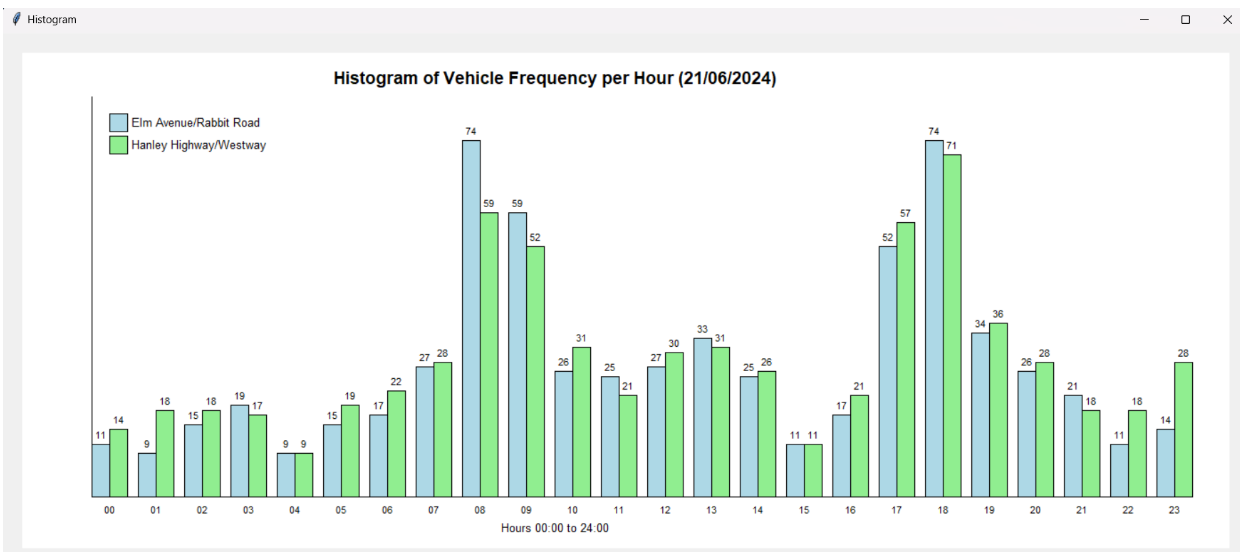

Figure 18 Screenshots in Histogram